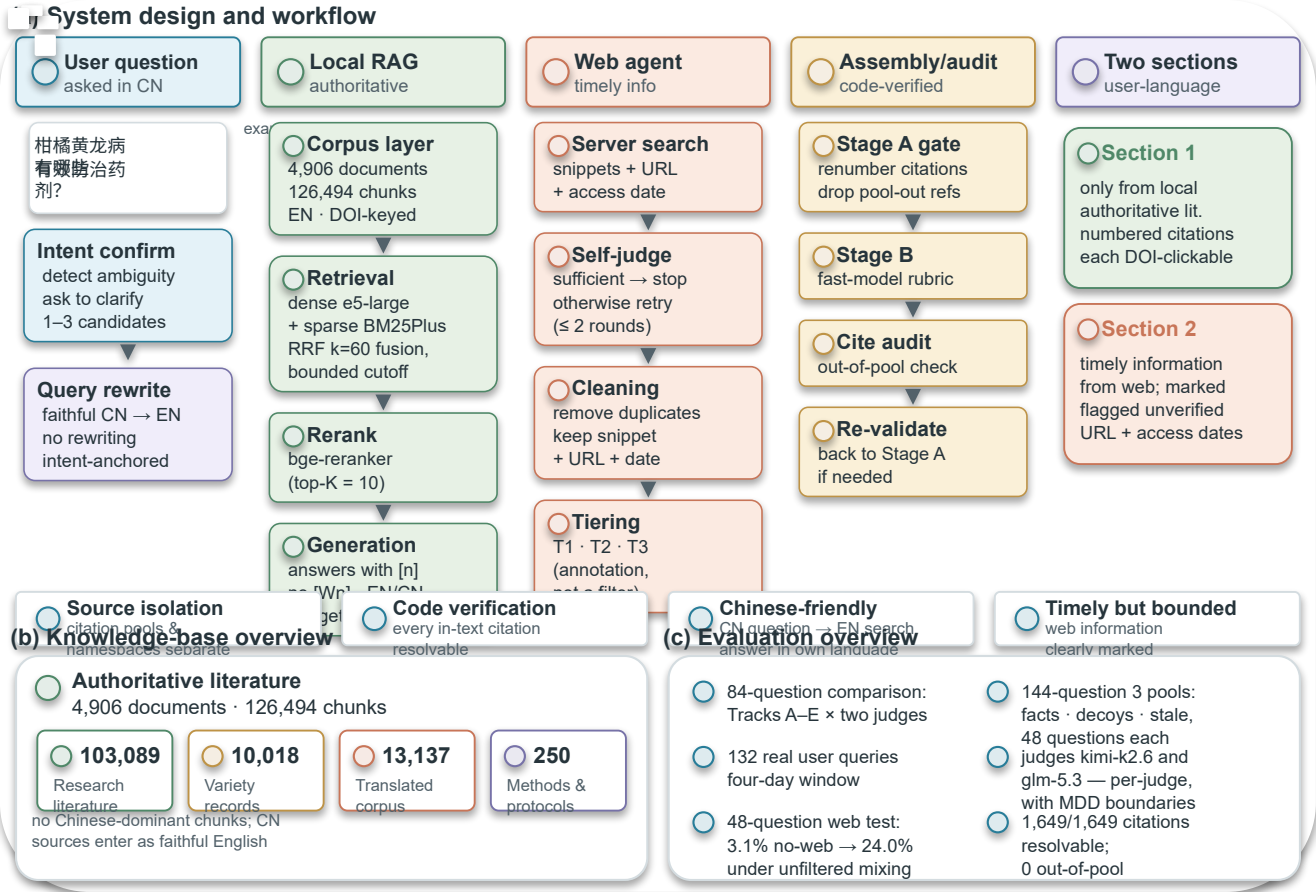

System design and workflow
User question
asked in CN
柑橘黄龙病
有哪些治药
剂?
Intent confirm
detect ambiguity
ask to clarify
1–3 candidates
Query rewrite
faithful CN → EN
no rewriting
intent-anchored
Local RAG
authoritative
Corpus layer
4,906 documents
126,494 chunks
EN · DOI-keyed
Retrieval
dense e5-large
+ sparse BM25Plus
RRF k=60 fusion,
bounded cutoff
Rerank
bge-reranker
(top-K = 10)
Generation
Web agent
timely info
Server search
snippets + URL
+ access date
Self-judge
sufficient → stop
otherwise retry
(≤ 2 rounds)
Cleaning
remove duplicates
keep snippet
+ URL + date
Tiering
T1 · T2 · T3
(annotation,
Assembly/audit
code-verified
Stage A gate
renumber citations
drop pool-out refs
Stage B
fast-model rubric
Cite audit
out-of-pool check
Re-validate
back to Stage A
if needed
Two sections
user-language
Section 1
only from local
authoritative lit.
numbered citations
each DOI-clickable
Section 2
timely information
from web; marked
flagged unverified
URL + access dates
Source isolation
citation pools &
Code verification
every in-text citation
resolvable
Chinese-friendly
CN question → EN search
Timely but bounded
web information
clearly marked
(b) Knowledge-base overview
Authoritative literature
4,906 documents · 126,494 chunks
103,089
Research literature
10,018
Variety records
13,137
Translated corpus
250
Methods & protocols
no Chinese-dominant chunks; CN
sources enter as faithful English
(c) Evaluation overview
84-question comparison:
Tracks A–E × two judges
132 real user queries
four-day window
48-question web test:
3.1% no-web → 24.0%
under unfiltered mixing
144-question 3 pools:
facts · decoys · stale,
48 questions each
judges kimi-k2.6 and
glm-5.3 — per-judge,
with MDD boundaries
1,649/1,649 citations
resolvable;
0 out-of-pool